TXM0008 - User and installation guide

UV-K5 FAGCI Community | OSFW-TXM0008 | 15 October 2026

This independent community build is based on FAGCI MOD v1.2.1 / DualTachyon. It includes all local additions from NAME001/SPC0002 through TXM0008. It is not an official FAGCI, Egzumer, IJV or Quansheng release.

The maintainer reported successful operation of TXM0008 on their radio on 15 October 2026. In addition, 135 automated tests passed against compiled ARM firmware and the CHIRP core with simulated peripherals. These are different forms of evidence; neither establishes compatibility with every hardware revision or a measured RF sensitivity improvement.

1. Files and compatibility

File	Purpose
firmware_TXM0008.bin	Raw firmware, 60,668 bytes. Use with a raw-compatible loader such as K5Prog.
firmware_TXM0008.packed.bin	Packed firmware, 60,686 bytes. Use only with a loader supporting that format.
uvk5_fagci_txm0008.py	Standalone CHIRP module for OSFW-TXM0008.
TXM0008_complete_package.zip	Firmware, CHIRP module, guides, K5Prog, sources, credits and licenses.
sursa_FAGCI_TXM0008_publicare.zip	Complete firmware source, dependencies, CHIRP module and tests; not a radio image.
K5Prog-v1.26-official-with-source.zip	Unmodified official OneOfEleven Windows tool, GPL license and corresponding source archive.
SHA256_PUBLICARE_TXM0008.txt	Checksums of the published assets.

This source targets the original DP32G030 platform. Compatibility with GD32/PY32 or other processor/board variants is not established. Check the hardware rather than relying on a similar model name. No EEPROM reset is required when updating from the earlier builds of this community project.

Firmware and EEPROM configuration are different: flashing the firmware updates the program; a CHIRP .img is a backup/configuration image. Keep each radio's own configuration and calibration. Do not replace them with someone else's dump.

2. Install with K5Prog on Windows

The bundle contains official **K5Prog Windows v1.26**, pinned to a published OneOfEleven source revision, together with its GPL license and corresponding source. It is not the separately modified IJV executable used by the maintainer. That local IJV executable was not redistributed because its exact corresponding source was not established. The included official executable has not been physically tested with this radio during packaging.

1. Extract the complete package to a local folder. Use the original programming cable.

2. Close CHIRP or other software using the same COM port.

3. For backup, turn the radio on normally. In K5Prog select the cable's COM port and connect.

4. Use **Read Configuration** and **Read Calibration**, keeping the saved files privately.

5. Disconnect in K5Prog and turn the radio off.

6. Hold **PTT** while powering on to enter the firmware-update mode. The flashlight LED lights.

7. Connect in K5Prog using the correct COM port, then choose **Write Firmware**.

8. Select **firmware_TXM0008.bin**. This is the raw firmware, not a source ZIP or CHIRP .img.

9. Wait for completion without interrupting the connection or power. Then switch the radio off and restart normally.

10. Check OSFW-TXM0008 using CHIRP identification or the build information, then check your channels and settings.

Raw and packed files are separate assets; do not rename one to pretend it is the other. The official K5Prog source describes support for both raw and packed firmware. The raw file above keeps the flashing workflow used in this project.

K5Prog upstream: https://github.com/OneOfEleven/k5prog-win

3. Main menu: the M key

Press M briefly from the main screen. Select a menu with UP/DOWN or its number, press M to edit, change the value, and press M to accept. EXIT cancels an unsaved edit; another EXIT closes the menu. MEM-CH, DEL-CH and RESET require an additional SURE? confirmation. Holding M does not replace a separate confirmation press.

Normal menu numbers 1-50 are retained. TX-CH is the new normal menu 51.

No.	Display	Function	Values or operation
01	**SQL**	Squelch threshold; 0 keeps reception open.	0-9
02	**STEP**	Tuning step. Changes through this menu are accepted only in VFO mode.	0.01 / 0.1 / 0.5 / 1 / 2.5 / 5 / 6.25 / 8.33 / 10 / 12.5 / 25 / 100 kHz
03	**TXP**	Transmit power for the active VFO/channel.	LOW / MID / HIGH
04	**R_DCS**	Receive DCS code.	OFF or normal/inverted DCS code
05	**R_CTCS**	Receive CTCSS tone.	OFF or one of 50 tones
06	**T_DCS**	Transmit DCS code.	OFF or normal/inverted DCS code
07	**T_CTCS**	Transmit CTCSS tone.	OFF or one of 50 tones
08	**SFT-D**	Transmit frequency offset direction.	OFF / + / -
09	**OFFSET**	RX-TX frequency offset, displayed in MHz.	Number keys or arrow keys
10	**W/N**	Channel filter bandwidth.	WIDE / NARROW
11	**SCR**	Channel scrambling, intended for FM.	OFF / 1-10; also depends on SCREN
12	**BCL**	Blocks transmission when the channel is busy.	OFF / ON
13	**MEM-CH**	Stores the current parameters in the selected memory slot. Requires SURE? confirmation.	CH-001 to CH-200
14	**SAVE**	Battery saving. SAVE here does not store a channel.	OFF / 1:1 / 1:2 / 1:3 / 1:4
15	**VOX**	Voice-activated transmission and trigger level.	OFF / 1-10
16	**BACKLI**	Display backlight duration.	OFF / 5 sec / 10 sec / 20 sec / 1 min / 2 min / ON
17	**DW**	Dual Watch: monitors both VFOs, with the associated A/B selection.	OFF / CHAN_A / CHAN_B
18	**WX**	Routes reception and transmission between VFOs A/B. In this source, it is the cross-band setting, although the display says WX.	OFF / CHAN_A / CHAN_B
19	**BEEP**	Keypad sounds.	OFF / ON
20	**TOT**	Maximum duration of a transmission.	OFF / 1-10 min
21	**VOICE**	Inherited voice announcement language selector. This source does not implement voice playback linked to this setting.	OFF / CHI / ENG
22	**SC-REV**	Scan behavior when a signal is detected: timed resume, resume after the carrier disappears, or stop.	TO / CO / SE
23	**MDF**	Memory channel display format.	FREQ / CHAN / NAME / NAME+F
24	**AUTOLK**	Automatic keypad lock.	OFF / ON
25	**S-ADD1**	Includes the channel in scan list 1.	OFF / ON
26	**S-ADD2**	Includes the channel in scan list 2.	OFF / ON
27	**STE**	Squelch tail elimination, where applicable.	OFF / ON

No.	Display	Function	Values or operation
28	**RP-STE**	Repeater tail elimination delay.	OFF / 1-10 x 100 ms
29	**MIC**	Transmit microphone sensitivity.	0-4
30	**1-CALL**	One-touch call channel, subsequently accessed with F → 9.	CH-001 to CH-200
31	**S-LIST**	Default memory channel scan list.	LIST1 / LIST2
32	**SLIST1**	Views list 1 and its priority channels. Does not edit priorities.	Channels in the list / NULL
33	**SLIST2**	Views list 2 and its priority channels. Does not edit priorities.	Channels in the list / NULL
34	**ANI-ID**	Displays your own DTMF identifier. Cannot be edited here.	View only
35	**UPCODE**	Displays the DTMF code sent at the beginning of transmission. Cannot be edited here.	View only
36	**DWCODE**	Displays the DTMF code sent at the end of transmission. Cannot be edited here.	View only
37	**D-ST**	Local monitoring of DTMF tones.	OFF / ON
38	**D-RSP**	Response to a decoded DTMF call.	NULL / RING / REPLY / BOTH
39	**D-HOLD**	DTMF call hold/automatic reset time.	5-60 sec
40	**D-PRE**	Delay before sending the DTMF code.	3-99 x 10 ms
41	**PTT-ID**	When the DTMF identifier is sent during PTT operation.	OFF / BOT at start / EOT at end / BOTH
42	**D-DCD**	Enables DTMF decoding on the channel.	OFF / ON
43	**D-LIST**	Selects a DTMF contact. Confirming a valid contact prepares it for a call.	1-16 / NULL for an invalid entry
44	**PONMSG**	Startup screen: full display, message, or battery voltage.	FULL / MSG / VOL
45	**ROGER**	End-of-transmission signal.	OFF / ROGER / MDC
46	**VOL**	Displays battery voltage. VOL in this menu is not a volume adjustment.	View only, in volts
47	**MODUL**	Basic modulation selector in the main menu.	OFF = FM / ON = AM; see modulation notes
48	**DEL-CH**	Deletes the selected channel from usable memory. Requires SURE? confirmation.	CH-001 to CH-200
49	**RESET**	Resets settings or performs a full reset. Requires SURE? confirmation and restarts the radio.	VFO / ALL
50	**NAME**	Views and edits the active memory channel name.	Up to 10 ASCII characters; see steps below
51	**TX-CH**	Persistent TX permission for the selected stored memory.	OFF = RX-only / ON = allowed subject to global limits

4. Edit a channel name: menu 50 NAME

Select an existing memory channel, then press M, 5, 0, M. The screen identifies the channel and shows the full name, independently of the MDF display setting.

Key	Action in NAME
UP / DOWN	Change the current character; hold to repeat.
F	Move the cursor right.
*	Move the cursor left.
0-9	Insert that digit and advance.
M	Save and leave editing.
EXIT	Cancel the unsaved changes.

Names use up to ten printable ASCII characters. Replace trailing characters with spaces to shorten a name; ten spaces clear it. NAME displays MR ONLY outside a valid memory. It updates only the name, preserving the RX profile and TX flag.

5. Save a frequency from VFO

Enter or tune the frequency. If you typed it manually, accept the frequency with M first. **Hold Side Key 2** in VFO to open MEM-CH at the first free memory slot. Select CH-001 through CH-200, press M for SURE?, then press M again to save. EXIT cancels. Changing the destination invalidates the previous confirmation.

Side Key 2 is the lower small side button. Its short action remains configurable; long-press VFO saving takes priority in VFO. Outside VFO, the configured long action remains available. Ordinary VFO parameters are saved through the existing memory serializer. An occupied destination is overwritten, with its existing TX block retained.

The ordinary menu route remains M -> 13 MEM-CH -> M -> destination -> M -> SURE? -> M. Menu 14 SAVE is battery saving, not channel storage. Some ordinary changes made while viewing an MR channel need an explicit MEM-CH save to persist. NAME, TX-CH and scan-list participation have their own persistence paths.

6. Spectrum: enter, listen and save

From the main screen press F, release it, then 5. Pressing PTT briefly on the graph enters fixed listening on the detected peak; release PTT. A later PTT press in fixed listening can transmit if the current transmit memory and global limits permit it. PTT is not the save button.

Press and release **Side Key 2** to open memory saving. From the graph, the peak frequency is captured; from fixed listening, the tuned frequency is captured. The first free slot is proposed. Choose a destination with UP/DOWN, press M for SAVE? or REPLACE?, then press M again to save. EXIT cancels from either stage. Changing the slot invalidates confirmation; held M cannot overwrite automatically. PTT is ignored in this save screen. Select the saved memory after leaving Spectrum.

Frequency, modulation, tuning step and Wide/Narrow filter are saved. A Spectrum 6.25 kHz filter is stored as Narrow because the channel format has only two bandwidth states. The new Spectrum memory has no tones or repeater offset, uses the active VFO's power, and belongs to both scan lists. An existing name and TX block are retained when overwriting an occupied slot; a new slot starts with a blank name.

In fixed listening, M cycles through LNAs, LNA, PGA, MIX, IF, DEV, CMP and MIC; UP/DOWN adjusts the selected register. EXIT first leaves register selection, then returns to the graph; EXIT from the graph leaves Spectrum. Key 0 cycles modulation.

7. Optional saved RX profile

Spectrum saving starts at **RX:AUTO**. While saving from **fixed listening only**, press **F** to choose RX:MANUAL; F toggles AUTO/MANUAL. The gain and IF values were captured when saving opened. Changing this mode or destination requires a new confirmation. From the scanning graph only AUTO is available.

- AUTO keeps normal operation, including the automatic AM controller.

- MANUAL stores the LNAs/LNA/MIX/PGA gain register and the IF register for that memory.

- MANUAL restores fixed gain on memory selection and restart; the AM controller does not override it.

- Resave the same slot as AUTO to remove its manual profile. NAME and TX-CH preserve the profile.

- DEV, CMP, MIC and the Spectrum squelch/RSSI threshold are not saved in the RX profile.

The profile uses the six reserved bytes after the ten-character name; it does not shorten the name. AUTO remains the default, particularly for changing aircraft signal levels. Fixed gain is an explicit option and does not guarantee improved reception.

8. Per-channel transmit control: menu 51 TX-CH

Select the stored channel used for TX, then press **M -> 5 -> 1 -> M**. Choose OFF for RX-only or ON to permit TX subject to global limits. M saves immediately; EXIT cancels. CH-xxx identifies the destination. Matching caches in both VFOs update immediately, and the flag persists after restart.

GLOBAL OFF means the global ALL_TX setting is DISABLE; selecting ON does not override it. MR ONLY means no valid stored memory is selected. This is a per-memory flag, not a frequency blacklist: the same frequency entered in VFO or another memory is independent. In Cross-band, the actual transmit memory determines the flag.

PTT, both side keys and F are ignored inside this editor. The destination is pinned on entry. Saving the same choice does not write EEPROM again. Only the eight-byte block containing the existing TX marker is written; no new EEPROM space is allocated.

Normal PTT, alarm/1750-Hz paths and Spectrum respect the block. Spectrum/VFO overwrites of an occupied blocked slot keep it blocked. Explicit TX-CH ON or CHIRP TX enabled clears the block. Deleting and recreating a memory resets its old properties. Firmware older than TXP0007 does not enforce this marker.

9. CHIRP: load the correct module

1. Save any existing unsaved CHIRP work.

2. Choose File -> Load Module and open uvk5_fagci_txm0008.py.

3. If that menu is missing, enable Help -> Developer Mode, restart, and load the module.

4. Choose Radio -> Download From Radio with your cable port, vendor Quansheng and model UV-K5 FAGCI TXM0008.

5. Keep the radio in normal mode, outside Spectrum. Save the new .img as your original backup.

6. Edit the new image, save it, and use Radio -> Upload To Radio with the same model.

An old IJV/TXP0007 tab is not converted by loading the new module. The driver accepts OSFW-TXM0008 exactly and requires its own download baseline before upload.

The table exposes frequency, name, FM/NFM/AM/NAM/USB, actual tuning step, tones, power and offset. NAM means narrow AM. Open the channel properties and **Extra** for TX enabled, bandwidth, scan lists, DTMF options, scrambling and the RX profile. Disabled TX is also represented as Duplex off. Rechecking TX enabled restores the stored offset/direction instead of replacing it with a fake offset.

MANUAL gain fields are LNAs/LNA/MIX/PGA indices and IF as a decimal register value. For AUTO memories they are proposed defaults for creating a manual profile, not live AGC measurements. BYP/RAW are selected in Extra; CHIRP represents them as FM/NFM in the main Mode column. Unrelated edits preserve these special modulations.

Settings include radio/side-key/scan options, startup messages, DTMF codes, 20 broadcast-FM presets and 16 contacts. Calibration, password and AES-key upload are protected. The fourteen internal VFO slots and unsaved Spectrum registers are not editable. Only edited bytes are merged into current EEPROM blocks; all conflicts are checked before writing, and written blocks are reread for verification. After changing a setting directly on the radio, download again before editing that same value in CHIRP. Keep native .img backups privately.

10. AM reception and the SOS flashlight

For normal AM reception, the firmware explicitly selects the fixed hardware gain slot while the inherited FAGCI software AM controller adjusts gain. Other normal modes use hardware automatic gain; Spectrum uses hardware AGC and restores it on exit. The FAGCI gain table/controller source was not replaced with Egzumer's. The owner reported improved AM reception during earlier testing; no laboratory sensitivity, dynamic-range or audio-quality figures are claimed.

The flashlight action cycles **steady -> repeated SOS -> off**. SOS is three dots, three dashes, three dots, then a pause. It restarts at the first dot when reactivated. The nominal Morse unit is 160 ms, with a 5.44-second complete cycle. Scheduling can affect real timings. As with the old blinking function, the separate Spectrum loop pauses the sequence until Spectrum exits. RX/TX status LEDs are unchanged.

11. Special menus 52-57

Power on while holding **PTT + Side Key 1**, the upper small button immediately below PTT. Release the buttons when the special menu appears. These settings are TX permissions and global scrambling, not AM sensitivity controls.

No.	Display	Function	Values
52	350TX	Configured 350 MHz TX permission.	OFF / ON
53	F-LOCK	Transmit band-limit profile.	OFF / FCC / CE / GB / LPD PMR
54	200TX	Configured 200 MHz TX permission.	OFF / ON
55	500TX	Configured 500 MHz TX permission.	OFF / ON
56	ALL_TX	Global transmission mode.	DEFAULT / ENABLE / DISABLE
57	SCREN	Global scrambling permission.	OFF / ON

Use transmit functions only where you are authorized. No global permission is changed automatically by this firmware update or by TX-CH.

12. Limitations and validation

Menu 47 MODUL selects only FM/AM. The five stored modulations can be cycled with F then 7, or key 0 in Spectrum. Ordinary MR parameter changes may need MEM-CH. STEP changes through the normal menu apply only in VFO; CHIRP can edit a memory's stored step. Voice playback is not implemented by the inherited VOICE setting.

Raw firmware occupies 60,668 of 61,440 FLASH bytes; 772 bytes remain. Packed size is 60,686 bytes. Memory fullness does not by itself slow menu navigation. TX-CH reuses the existing EEPROM marker and avoids unchanged writes.

135 automated tests passed: 70 firmware regressions and 65 CHIRP/ARM/TX/menu tests. The owner separately confirmed successful TXM0008 operation on 15 October 2026. Exact board revision and exhaustive per-feature/hardware coverage were not provided. The packaged official K5Prog v1.26 was checked for provenance, version, license and matching source archive; it was not executed or used to flash a radio by the assistant.

13. Source, credits and licenses

Build the complete source with bash build-firmware.sh using ARM GCC and Python crcmod. The actual CMSIS_5/printf dependency snapshot is included. Earlier missing dependencies were recovered from the local Egzumer 0.22 archive; see dependency provenance and retained license notices. Build dates/toolchains may change hashes. Published SHA256 values identify the distributed binaries.

Base projects: FAGCI, DualTachyon; AM-fix contributors include OneOfEleven. Egzumer 0.22 was used as a comparison reference and dependency archive. CMSIS remains attributed to ARM; printf to Marco Paland and contributors. CHIRP transport credits: Jacek Lipkowski and Dan Smith; the supplied IJV adaptation also credits Julian Lilov LZ1JDL and Francesco IK8JHL. K5Prog Windows is by OneOfEleven, based on Jacek Lipkowski's original k5prog.

Firmware: Apache-2.0 with dependency notices. CHIRP module and K5Prog: their included GPL notices and source apply separately. Local development and documentation were prepared with Codex assistance. Original authors are credited; no endorsement by upstream projects or Quansheng is implied.